Roadway Infrastrastructure Economic Resilience

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Problem Statement

Current pavement selection decisions fail to evaluate pavement alternatives as a long-term system with external interactions. The investment decision has financial impacts that compound over decades through maintenance cycles, market exposure, and infrastructure disruptions. This project aims to improve the economic resilience of the City of Charlotte's roadway infrastructure.

Objectives

1. Quantify the pavement life cycle costs including initial costs, maintenance, major repairs and residual value.
2. Apply capital costing methods, including NPV, to support comparison of pavement alternatives under different economic and operational scenarios.
3. Evaluate uncertainty and decision robustness using scenario based and sensitivity analysis.
4. Quantify societal and sustainability impacts associated with pavement alternatives.
5. Provide recommendations for Charlotte infrastructure decisions on future pavement selection and financial planning.
6. Apply the analysis to a representative Charlotte roadway case study to compare direct agency costs with broader societal and environmental impacts.

NPV Estimation

- MARR: 4.1%
  - 30 year nominal interest rate on treasury notes (Office of Management and Budget, 2026)
- Study Life: 35 years (FHWA, 1998)
- PCC Useful Life ≅ 40 Years (Robbins & Tran, 2018)
- HMA Useful Life ≅ 35 Years
- Residual Value = [(Expected Life of Rehab - Time Since Rehab) / Expected Life of Rehab] × Cost of Rehab
  - PCC: 15.38 $/SY
  - HMA: 17.68 $/SY

Alternate	Initial Cost ($/SY)	O&M #1	Cost ($/SY)	Interval	O&M #2	Cost ($/SY)	Interval
PCC	123.13	Joint Sealing	0.2	Every 5 Years	Diamond Grinding	3.86	Initial at 12 years, then every 10 years
HMA	62.46	Crack Repair	0.7	Every 2 Years	1.5" Mill and Repave	35.36	Every 10 years

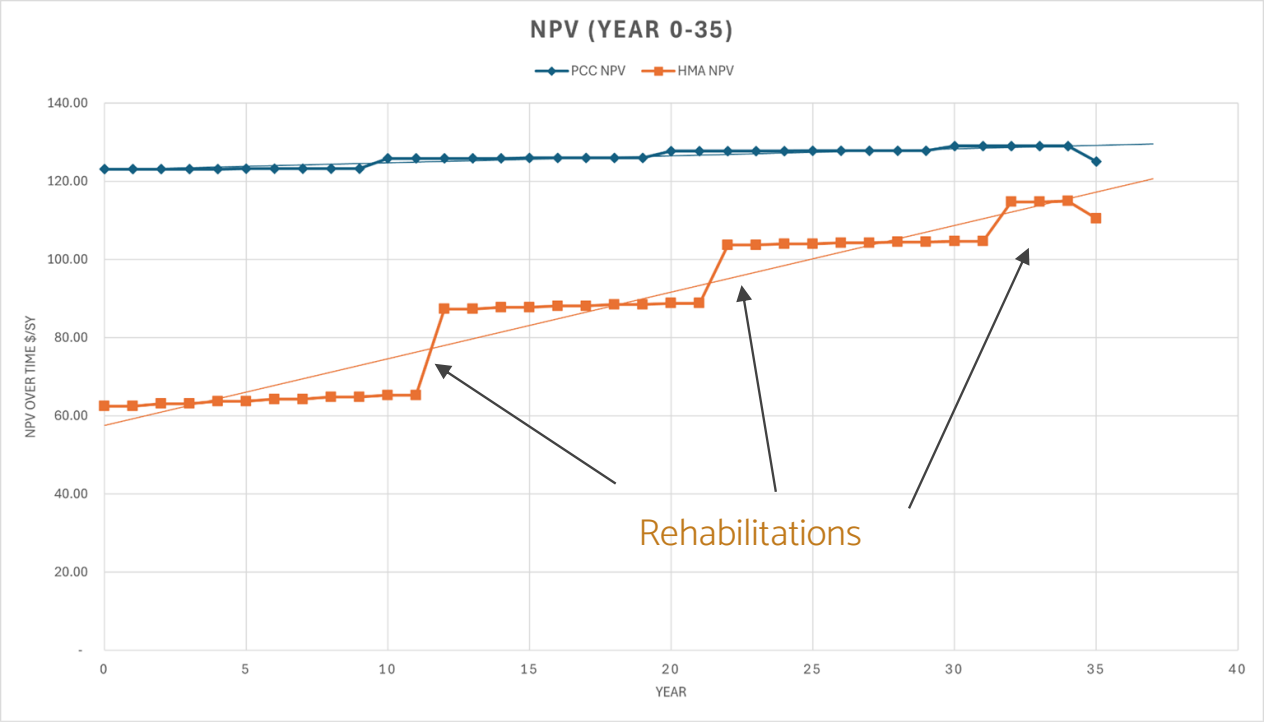
NPV and Annual Worth

NPV:

- HMA: 110.14 $/SY
- PCC: 125.04 $/SY

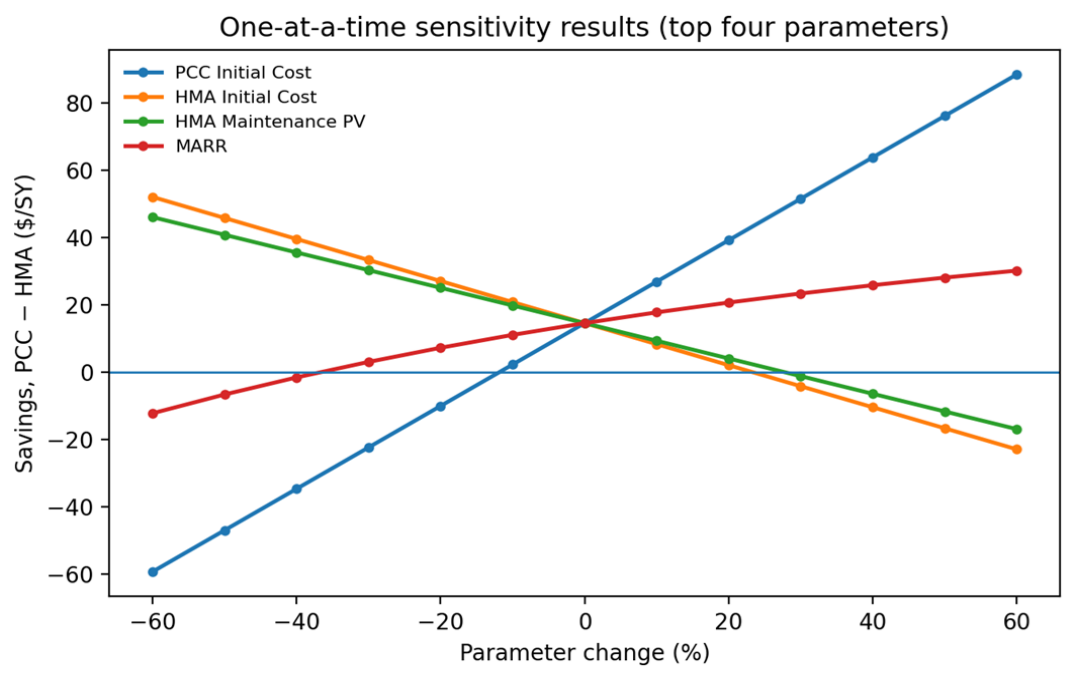
Annual Worth:

- HMA 6.00: $/SY
- PCC: 6.79 $/SY



NPV Sensitivity Analysis

Base Case and One at a time - Sensitivity Analysis



Base Case

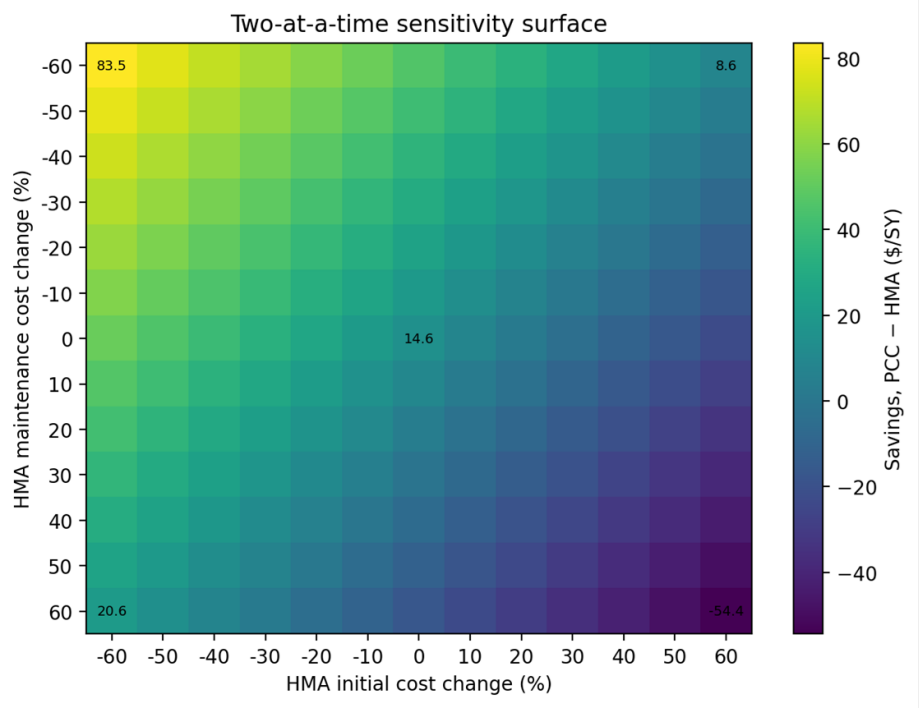
PCC = $125.04/SY
HMA = $110.46/SY
Savings = +$14.59/SY - HMA Preferred

	6 Parameters	Slope	Reversal
1	PCC Init.	123.1	-11.85%
2	HMA Init.	62.5	+23.35%
3	HMA Maint.	52.5	+27.80%
4	MARR	35.3	-36.65%
5	PCC Maint.	5.9	No rev.
6	Salvage Value	0.5	No rev.

The decision is mainly driven by PCC initial cost, HMA initial cost, HMA maintenance PV, and MARR.

NPV Sensitivity Analysis

Two at A Time - Sensitivity Analysis



Base point

Center cell = +$14.59/SY
HMA remains preferred in the submitted base case.

High-cost region

As both HMA cost drivers rise together, savings shrink and turn negative.
That is where PCC becomes cheaper than HMA.

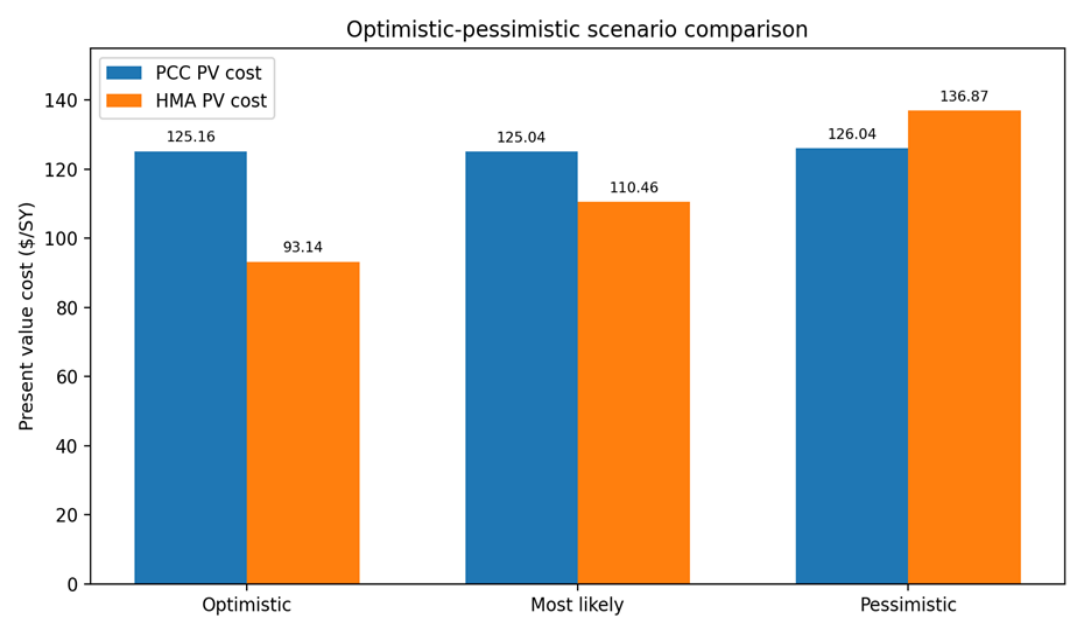
Decision meaning

This method shows that the base-case result is robust under moderate combined changes, but not under simultaneous adverse HMA cost escalation.

Two-at-a-time analysis shows where combined HMA cost increases can reverse the recommendation.

NPV Sensitivity Analysis

Optimistic-Pessimistic Analysis



Scenario	PCC	HMA	Savings	Preferred
Optimistic	125.16	93.14	+32.02	HMA
Most Likely	125.04	110.46	+14.59	HMA
Pessimistic	126.04	136.87	-10.83	PCC
Expected	125.32	112.73	+12.59	HMA

Why this matters for the decision

1. Confirms HMA under base and expected conditions.
2. Shows PCC becomes preferable only in the pessimistic case.
3. Gives clear thresholds to monitor before changing the recommendation.

HMA remains the lower life-cycle cost option under expected conditions; shift toward PCC only if adverse HMA cost conditions occur together.

Scenario Analysis: Usage-Based Rehabilitation

Traffic loading determines rehabilitation frequency. More rehab events means higher total cost for HMA.

Rehabilitation Intervals

Scenario	Prob.	HMA yrs	PCC yrs	HMA Events
High Use	25%	8	18	**4**
Average Use	50%	11	24	**3**
Low Use	25%	14	30	**2**
Wtd. Avg	**100 %**	**11.0**	**24.0**	**--**

PCC diamond grinding intervals are 2 to 3x longer than HMA overlay intervals across all scenarios.

35-Year NPV Results ($/SY)

Scenario	HMA	PCC	Difference	Winner
High Use	$137.49	$125.62	**($11.87)**	**PCC**
Average Use	$116.48	$125.22	**$8.74**	**HMA**
Low Use	$101.11	$124.91	**$23.80**	**HMA**
Prob.-Wt. Exp.	**$117.89**	**$125.24**	**$7.35**	**HMA**

HMA is the expected lower-cost alternative by $7.35/SY. Advantage flips to PCC if high-use conditions develop.

Scenario Analysis: Oil Market Risk

Asphalt binder is petroleum-derived. Every future HMA resurfacing event carries oil price risk that PCC does not share.

PCC NPV is constant across all oil conditions at $125.22/SY because its cost structure contains no petroleum component.

Oil Market Condition	Probability	HMA Multiplier	HMA NPV ($/SY)	PCC NPV ($/SY)	Winner
High Cost (+100% of avg)	25%	**1.70x**	**$193.52**	$125.22	**PCC**
Average Cost (Base)	50%	1.00x	$116.48	$125.22	**HMA**
Low Cost (-50% of avg)	25%	**0.50x**	**$61.45**	$125.22	**HMA**
Prob.-Weighted Expected	**100%**	**1.05x**	**$121.98**	**$125.22**	**HMA**

$132/SY
HMA NPV swing across oil conditions

$68.30/SY
PCC advantage under high oil prices

$3.24/SY
HMA expected advantage (prob.-wtd.)

Scenario Analysis: Combined Findings and Recommendation

Both scenario frameworks evaluated. HMA is the expected lower-cost alternative across usage and oil market conditions.

Expected NPV Summary ($/SY)			
Scenario Framework	**HMA Expected**	**PCC Expected**	**HMA Advantage**
Usage Scenarios	$117.89	$125.24	**$7.35**
Oil Market Scenarios	$121.98	$125.22	**$3.24**

Recommendation holds under two conditions:

1. Traffic volumes remain near average projections (not high-use corridor development)
2. Petroleum prices do not sustain a significant spike above historical averages

RECOMMENDATION

HMA (Asphalt)

Expected lower-cost alternative across both scenario frameworks

Usage Scenarios

HMA $117.89 vs PCC $125.24

HMA saves $7.35/SY

Oil Market Scenarios

HMA $121.98 vs PCC $125.22

HMA saves $3.24/SY

Societal Benefits

Maintenance and rehabilitation activities create work zones, reducing roadway capacity and increasing travel time for users. Societal impacts can be quantified using user delay costs (U.S. Department of Transportation, 2025), calculated as: Total Delay Hours × Value of Time ($/hour).

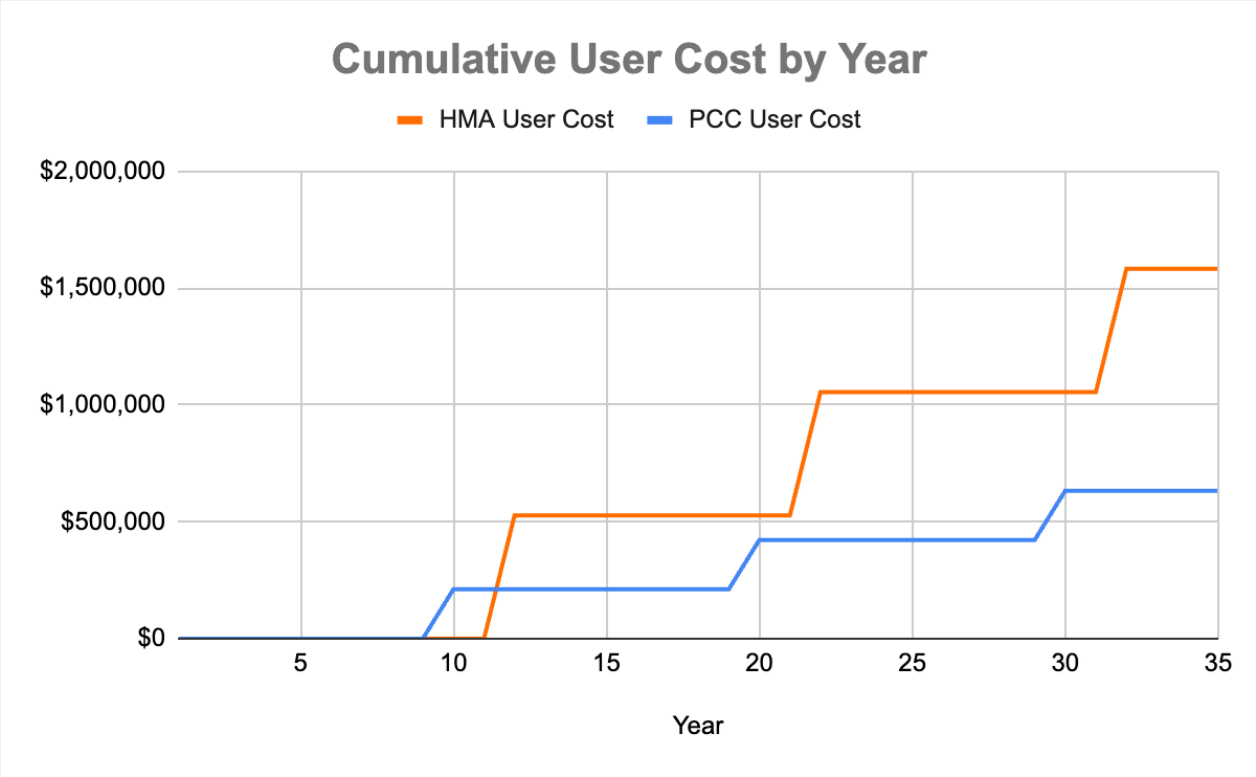
Average Daily Traffic (ADT)	20,000 vehicles/day (NCDOT, n.d.)
Delay per vehicle	15 minutes (0.25 hours)
Duration of work	Asphalt - 5 days Concrete - 2 days
Value of time	$21.10/hour (U.S. Department of Transportation, 2025)

Asphalt User Cost per Event
$=20{,}000\times0.25\times5\times21.10=\$527{,}500$

Concrete User Cost per Event
$=20{,}000\times0.25\times2\times21.10=\$211{,}000$

*User costs are estimated using representative traffic volumes (NCDOT), delay durations, and USDOT value of time guidance. Maintenance timing and work durations reflect typical pavement practices and are used for scenario-based comparison.

Societal Benefits



Sustainability

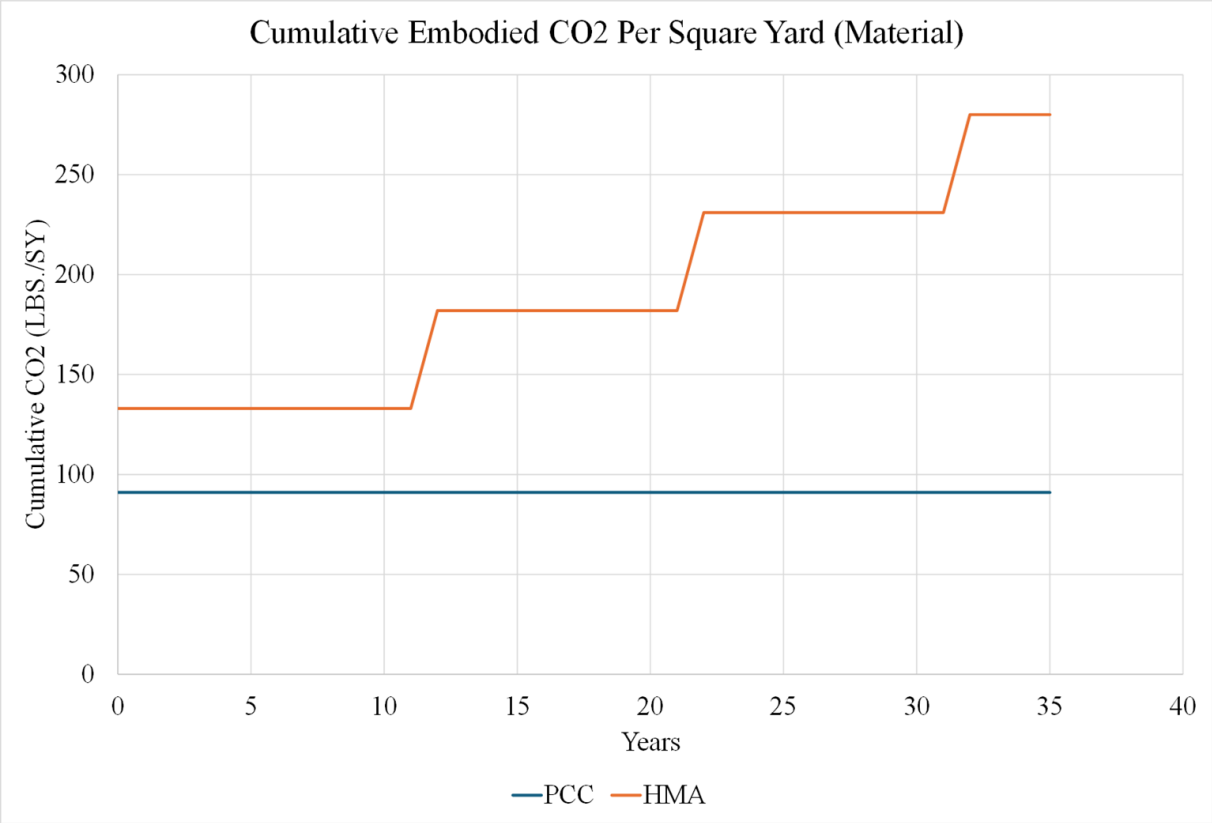
Factors :
Initial materials embodiment
Maintenance & rehabilitation activities
Frequency of maintenance cycles
Cumulative lifecycle CO_2 emissions

HMA total emissions ≈ 280 Lbs. CO_2 per SY over 35 years (Gruber, M. R., & Hofko, B., 2023)
PCC total emissions ≈ 95 Lbs. CO_2 per SY over 35 years (City of Portland, 2022)
Higher emissions in HMA are driven by frequent resurfacing cycles (Interagency Working Group on Social Cost of Greenhouse Gases, United States Government. 2016)

Lifetime Embodied CO2 and Environmental Cost:

Alternate	CO2 (Lbs/SY)	Cost ($/SY)
HMA	275	$13.5
PCC	100	$5

Sustainability



AI in Pavement Lifecycle Analysis

- Improves cost forecasting using NCDOT bid data
- Estimates maintenance timing based on pavement conditions
- Enhances risk analysis by capturing variability beyond predefined scenarios
- Reduces uncertainty in LCCA inputs
- Supports optimal maintenance planning to balance cost and sustainability

Case Study - Bryant Farms Road Extension

- Bryant Farms Road Extension: 1.2 mile planned road expansion in Charlotte, NC
- Estimated 25,344 SY of Pavement
- Expanded NPV calculated utilizing societal cost, carbon cost, and direct NPV under standard conditions

Bryant Farms Road Expanded NPV Results				
Cost Category	HMA ($/SY)	PCC ($/SY)	HMA Total Cost	PCC Total Cost
Direct NPV	$110.46	$125.04	$2,799,498	$3,168,014
Embodied carbon cost	$14	$5	$348,480	$126,720
User delay cost	$27	$12	$689,357	$298,794
Expanded NPV	**$151**	**$142**	**$3,837,335**	**$3,593,528**

Case Study - Recommendation

- Case Study Findings - Expanded NPV
  - PCC: $3,593,528
  - HMA: $3,837,335
- In Expanded NPV **PCC has a savings of nearly $250,000** over the project lifetime
- Applied scenario analysis under high use and high oil markets conditions
  - PCC offers over **$1.97 million in savings over** HMA over the lifetime in unfavorable conditions

Recommendation → PCC!

Conclusion

- Direct NPV analysis shows that HMA has 13% lower construction and maintenance costs than PCC
- Further systems analysis suggest that **PCC has a 7%** reduction of total related costs over HMA
- Overall, PCC provides greater long-term resilience when broader economic, societal, and environmental impacts are considered

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